

Day-6 Wireless Standard & Configuration

Task 1 :- Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

ANS :-

Add Router : Drag a Wireless Router (e.g., Home Router / WRT300N) into Packet Tracer.

Check SSID & Channel : Click Home Router → GUI tab → Wireless sub-tab:

- Default SSID: Default
- Default Channel: 1 or 6 (2.4 GHz band)

Connect Laptop / Smartphone:

- Drag a Laptop → double-click it → Physical tab → turn off power → swap Ethernet module for WPC300N wireless card → turn power back on.
- Go to Desktop tab → PC Wireless → Connect → select Default SSID → click Connect.

Verify : Wireless dotted lines appear between the Laptop and Home Router → Connected!

Task 2 :- Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks.

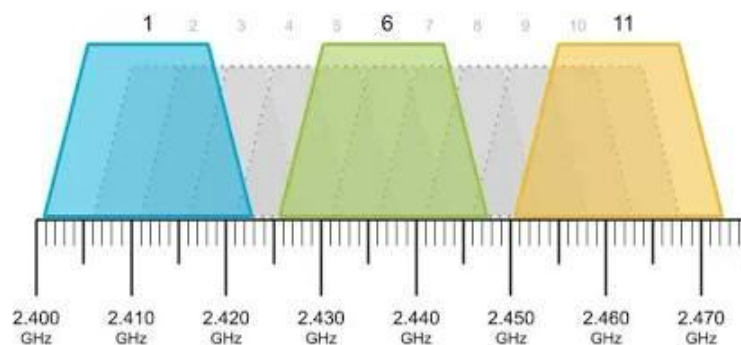
ANS :-

Change Wi-Fi Channel:

- Open Router GUI → Wireless tab.
- Change Standard Channel from Auto to 1, 6, or 11.
- Click Save Settings.

Check Interference on Phone:

- Open a Wi-Fi Analyzer app on your phone.
- Look at the 2.4 GHz Channel Graph.



Verify:

- Check if your chosen channel (1, 6, or 11) has fewer overlapping Wi-Fi signals from neighbors → Less interference = faster, more stable connection!

Task 3 :- Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

ANS :-

1. Set Password on Router :

- Open Router → GUI tab → Wireless → Wireless Security.
- Set Security Mode to WPA2 Personal.
- Type Passphrase: StrongPass123! → Click Save Settings.

2. Connect Device (PC / Laptop) :

- Open PC → Desktop → PC Wireless → Connect tab.
- Select network SSID → click Connect.
- Enter password StrongPass123! → click Connect.



Verify:

- Dotted wireless line appears → Status: Connected → Security Active!

Task 4 :- Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result.

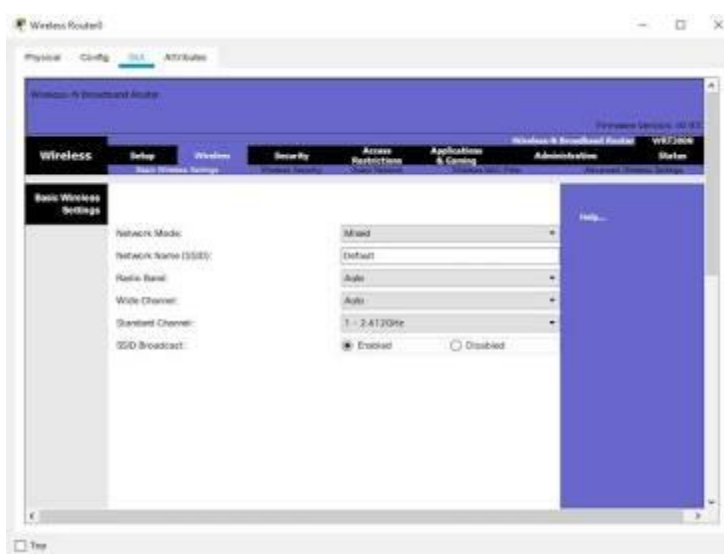
ANS :-

1. Get Device MAC Address :

- Open Laptop 1 → Desktop → Command Prompt → run ipconfig /all.
- Copy the Physical Address (MAC address, e.g., 0001.6311.AAAA).

2. Enable MAC Filtering on Router :

- Open Router → GUI → Wireless → Wireless MAC Filter.
- Select Enabled and pick Permit PCs listed below to access the wireless network.
- Enter Laptop 1's MAC address in the list → click Save Settings.



Test Both Devices (Results) :

- Laptop 1 (Allowed MAC): Connects successfully → Connected!
- Laptop 2 (Not in list): Connection fails even with correct Wi-Fi password → Access Denied!

Task 5 :- Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.

ANS :-

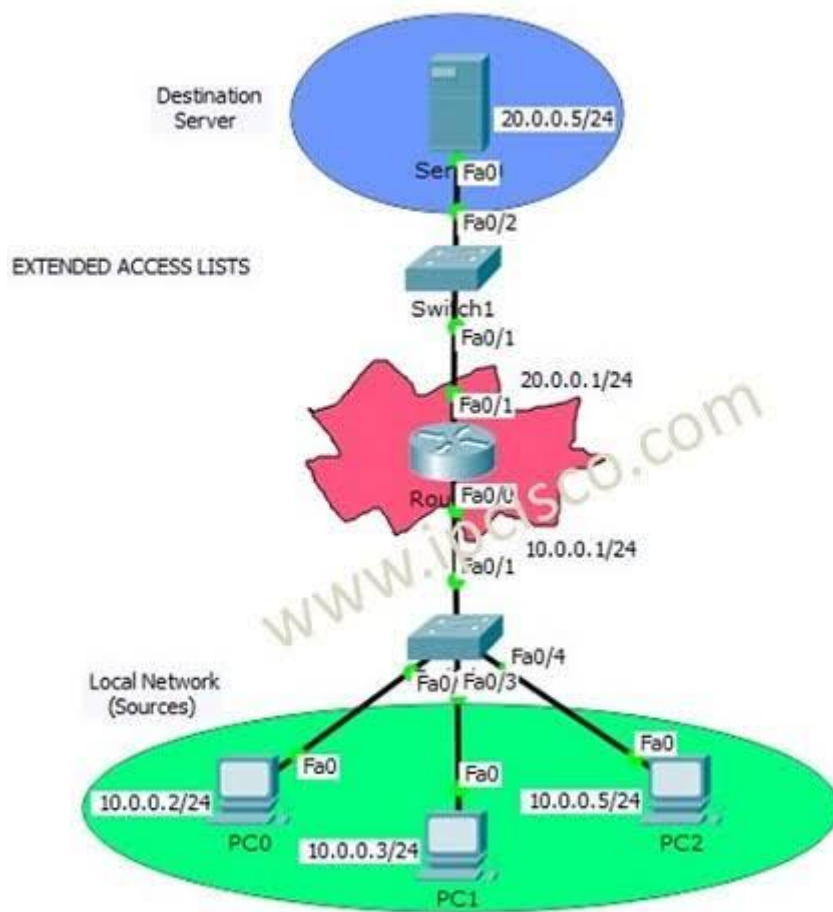
1. **Find Web Server IP** : Ping server URL (e.g., [youtube.com](https://www.youtube.com) → IP 20.0.0.5).

2. Configure Extended ACL on Router CLI :

```
enable
configure terminal
access-list 100 deny tcp any host 20.0.0.5 eq 80
access-list 100 permit ip any any
```

3. Apply ACL to Router Interface :

```
interface gi0/0/0  
ip access-group 100 in
```



Test & Verify :

- Open Web Browser on connected PC/Laptop → navigate to 20.0.0.5 or www.youtube.com.
- Result : "Request Timed Out" → Website Blocked!

